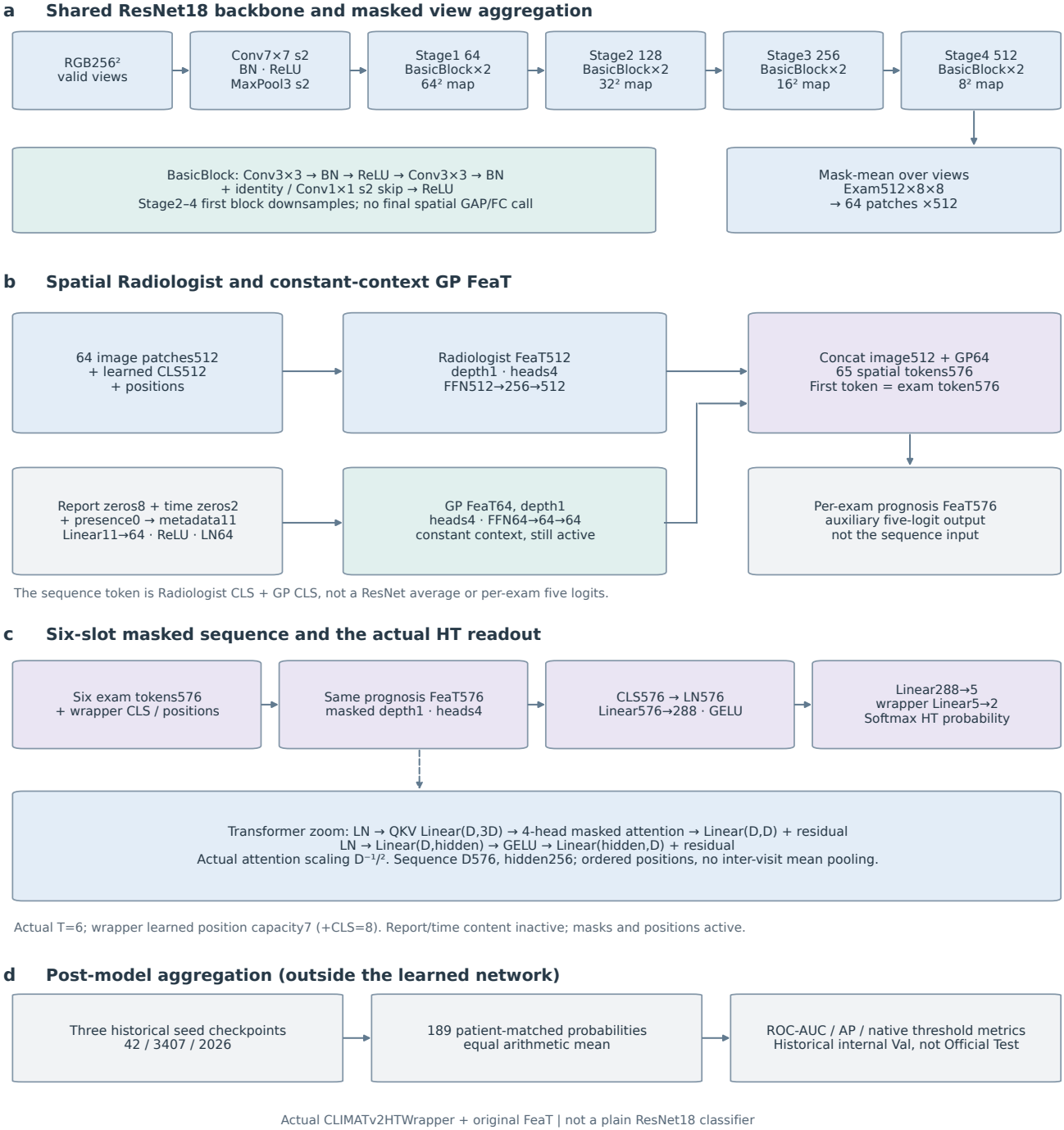
a
Shared ResNet18 backbone and masked view aggregation
RGB256²
valid views
Conv7×7 s2
BN · ReLU
MaxPool3 s2
Stage1 64
BasicBlock×2
64² map
Stage2 128
BasicBlock×2
32² map
Stage3 256
BasicBlock×2
16² map
Stage4 512
BasicBlock×2
8² map
BasicBlock: Conv3×3 → BN → ReLU → Conv3×3 → BN
+ identity / Conv1×1 s2 skip → ReLU
Stage2–4 first block downsamples; no final spatial GAP/FC call
Mask-mean over views
Exam512×8×8
→ 64 patches ×512
b
Spatial Radiologist and constant-context GP FeaT
64 image patches512
+ learned CLS512
+ positions
Radiologist FeaT512
depth1 · heads4
FFN512→256→512
Concat image512 + GP64
65 spatial tokens576
First token = exam token576
Report zeros8 + time zeros2
+ presence0 → metadata11
Linear11→64 · ReLU · LN64
GP FeaT64, depth1
heads4 · FFN64→64→64
constant context, still active
Per-exam prognosis FeaT576
auxiliary five-logit output
not the sequence input
The sequence token is Radiologist CLS + GP CLS, not a ResNet average or per-exam five logits.
c
Six-slot masked sequence and the actual HT readout
Six exam tokens576
+ wrapper CLS / positions
Same prognosis FeaT576
masked depth1 · heads4
CLS576 → LN576
Linear576→288 · GELU
Linear288→5
wrapper Linear5→2
Softmax HT probability
Transformer zoom: LN → QKV Linear(D,3D) → 4-head masked attention → Linear(D,D) + residual
LN → Linear(D,hidden) → GELU → Linear(hidden,D) + residual
Actual attention scaling D⁻¹/². Sequence D576, hidden256; ordered positions, no inter-visit mean pooling.
Actual T=6; wrapper learned position capacity7 (+CLS=8). Report/time content inactive; masks and positions active.
d
Post-model aggregation (outside the learned network)
Three historical seed checkpoints
42 / 3407 / 2026
189 patient-matched probabilities
equal arithmetic mean
ROC-AUC / AP / native threshold metrics
Historical internal Val, not Official Test
Actual CLIMATv2HTWrapper + original FeaT | not a plain ResNet18 classifier